

MOHAMMED SAMI JAITAPKAR

Graduate Mechanical Engineer | CAD • CFD • FEA • Build & Test

Portfolio: mosami10.github.io | LinkedIn: linkedin.com/in/sami-jaitapkar-5805012b0

Chelmsford, Essex | 07438 984876 | samijaitapkar10@gmail.com

PROFILE

Mechanical engineering graduate with five years of formal training: BEng (Hons) 2:1, Anglia Ruskin University (2026), built on a three-year Diploma in Mechanical Engineering (83%, Distinction). Works across the full loop of Inventor CAD, ANSYS CFD/FEA and physical build and test. Individual dissertation compared Formula 1 brake cooling ducts with conjugate heat transfer (0.82% GCI); coursework includes a BAE Systems live brief taken from FEA to additive manufacture and an HVAC duct redesign that cut pressure drop by 27%. Targeting graduate mechanical engineering roles.

EDUCATION

BEng (Hons) Mechanical Engineering, Upper Second Class (2:1)

2024 – 2026

Anglia Ruskin University, Chelmsford

- Dissertation: Design and Numerical Investigation of a Formula 1 Brake Cooling System (CFD and conjugate heat transfer).
- Key modules: Materials Science, Solid Mechanics & Vibration (73%); CAE & Manufacturing Project (70%); Advanced Structural Analysis & Design (67%); CFD & MATLAB/Simulink; Fluid Mechanics & Heat Transfer; Automation & Control; Advanced Engineering Mathematics; Design, Project Management & Quality Engineering.

Diploma in Mechanical Engineering, 83.13% (First Class with Distinction)

2019 – 2022

Mohammed Haji Saboo Siddik Polytechnic, Mumbai (three-year full-time technical diploma)

- Machine design, theory of machines, thermal engineering, fluid mechanics, strength of materials, engineering drawing, CAD and solid modelling, additive manufacturing. Final project: voice-based lift control system.

ENGINEERING PROJECTS

Formula 1 Brake Cooling Duct: CFD and Conjugate Heat Transfer

Individual dissertation, 2026

- Set out to answer a design question: which duct inlet removes the most heat from a 278 mm, 24-vane carbon-carbon brake disc at 1,500 rpm and 20 m/s? Compared an unducted disc, a three-outlet dome duct (A) and a single-outlet bell-mouth scoop (B), modelled in Autodesk Inventor.
- Ranked designs by heat transfer coefficient using isothermal moving-wall runs in ANSYS Fluent 2025 R2 (k- ω SST, pressure-based coupled solver), then ran staged conjugate heat transfer to resolve the temperature field inside the solid disc.
- Verified the solution with a three-mesh Richardson Grid Convergence Index of 0.82% on a 9.7M-cell mesh (published benchmark 1.40%), y^+ checks, and meshes built in both ANSYS Meshing and Fluent Meshing.
- Result: Duct B cooled the disc 109 K more than Duct A while carrying 65% less mass flow, traced to an 88.7% higher inlet Reynolds number on a 30.2 mm hydraulic diameter. CHT resolved a 712 K peak disc temperature and showed thermal runaway on the unducted disc.

Automotive HVAC Duct: CFD Optimisation and Cabin Thermal Model

CFD & Systems Engineering project

- Ran six CAD-CFD iterations: rapid screening in ANSYS Discovery, then detailed ANSYS Fluent runs on a poly-hexcore mesh, reworking the duct routing each loop.
- Cut system pressure drop by 27%, from ~199 Pa to ~145 Pa (final Design 4.2), while bringing the outlet mass-flow split to near-equal (~50/50) between the two vents.
- Reported the open trade-off honestly: peak velocity (~26 m/s) stayed above its limit and was set as the constraint for the next iteration.
- Coupled the CFD results to a Simulink/Simscape cabin heating model to check system-level thermal behaviour.

SPIDR-Jet Hybrid Drone Chassis

BAE Systems live brief

- Took a jet-powered hybrid drone chassis from CAD through four ANSYS Mechanical FEA iterations under weight and impact load cases.
- Rejected a full-lattice PLA concept that failed structural checks and selected a selective-lattice PETG chassis: 13.8% mass reduction to 93.48 g at a peak von Mises stress of 23.4 MPa.
- Checked manufacturability end to end: ANSYS Additive Print, Bambu Studio FDM slicing, then Tinius Olsen compression testing of printed parts.
- Modelled production in Witness Horizon discrete event simulation, scaling throughput from ~42 to ~500 units a month across printer-count and shift scenarios.

Leaf Spring: FEA, Materials Selection and Lab Validation

Module mark 73%

- Steel vs CFRP leaf spring in Inventor and ANSYS; CFRP met target stiffness at ~80% lower density; FEA checked against tensile and vibration testing.

NREL 5MW Wind Turbine Blade Fatigue Life

Individual report

- QBlade LLT + QFEM at six operating points; Weibull loading, Goodman-corrected S-N and Miner's rule to find the fatigue-governing case and estimate blade life.

Mechatronic Strawberry Cutter: Working Prototype

Design, build and test

- Inventor CAD, Multisim circuits and Arduino control; manufactured and commissioned in the workshop and tested through full cutting cycles.

TECHNICAL SKILLS

CAD & Design: Autodesk Inventor (part and assembly modelling, 2D engineering drawings), design for manufacture and assembly (DfMA), design for additive manufacture, lattice structures, Bambu Studio (FDM print planning, PETG/PLA)

CFD & FEA: ANSYS Fluent 2025 R2, Fluent Meshing (poly-hexcore, watertight geometry), ANSYS Meshing, ANSYS Discovery, ANSYS Mechanical / Static Structural, ANSYS Additive Print, QBlade (BEM/LLT + QFEM); k- ω SST, conjugate heat transfer, MRF, mesh independence, Richardson GCI, y+ checks

Thermal & Fluids: Heat transfer coefficient analysis, internal duct flow, pressure drop and flow balancing, thermal runaway assessment, fatigue (Goodman, Miner's rule), creep (Larson-Miller), fracture mechanics, NDT selection (PAUT)

Controls & MATLAB: MATLAB, Simulink, Simscape thermal networks, transfer functions, root locus, PI/PID tuning, Bode analysis, finite difference heat conduction solver, Arduino, Multisim

Manufacturing & Engineering: Workshop fabrication and assembly, FDM additive manufacturing, tensile, compression (Tinius Olsen) and vibration testing, Witness Horizon discrete event simulation, quality engineering, technical reports and presentations, Microsoft Excel

Approach: I can make it happen.

WORK EXPERIENCE

Warehouse Operative, First Call Agency, placed at DPD UK, Chelmsford

Oct 2025 – Present

- Sort, scan, load and stage parcels for next-day delivery routes to hourly throughput targets.

Front of House / Floor Team Member, The Courtyard

Feb 2026 – Present

- Restaurant, bar, live-event and weekend nightclub shifts, held alongside the warehouse role.

Customer Service Assistant, John Lewis Partnership, Chelmsford

Nov 2023 – Oct 2025

- Front-of-house service across multiple departments; awarded a Gold Star by management for helpfulness.

ACHIEVEMENTS & ADDITIONAL INFORMATION

- Diploma in Mechanical Engineering awarded First Class with Distinction (83.13%).
- Produced a BAE Systems-facing design pitch and ~4,000-word report for a university live brief.
- Gold Star award for helpfulness, John Lewis Partnership.
- Portfolio: nine engineering case studies with CAD, meshes, simulation results and conclusions at mosami10.github.io.
- Languages: English (native or bilingual); Hindi, Marathi and Urdu (professional working); Gujarati (limited working).